THE DERIVATIVE AS A FUNCTION

Darboux's Theorem

A derivative has the intermediate value property, even when it is not continuous.

THE THEOREM SAYS

If f is differentiable on an interval and f' takes the values A and B there, then f' takes every value between A and B somewhere on that interval.

IT DOES NOT SAY

~~If f is differentiable everywhere, then f' is continuous.~~

A DERIVATIVE THAT IS NOT CONTINUOUS

WRONG WORKING

f is differentiable at every real number, so f' must be continuous.

RIGHT WORKING

$f(x) = x^2 \sin(1/x)$ for $x \neq 0$, $f(0) = 0$, is differentiable everywhere, yet f' oscillates near the origin and is discontinuous there.

No jumps: the only discontinuities a derivative can have are oscillatory ones. Functions whose derivative is continuous everywhere form the class C^1.